

UNITY EDITOR & RUNTIME EXTENSION

CAD Importer

Import CAD models into Unity for robotics simulation and digital twins — STL, PLY, OBJ, STEP and IGES, with automatic LODs, simplified physics colliders and a runtime loading API.

Document: User Manual & Reference

Version: 1.0.0

Requires: Unity 6000.0 or newer · Universal Render Pipeline recommended

Package: com.motawea.cad-importer · github.com/Motawe3/CADImporter

Support: GitHub Issues · mohammed.motawea90@gmail.com

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1. Overview

CAD Importer turns mechanical CAD data into game-ready Unity content. It was designed for robotics simulation and digital-twin projects, where models arrive as engineering exports (millimeter units, Z-up axes, extremely dense tessellation, no game-oriented optimization) and must run at interactive frame rates alongside physics.

Every import automatically:

- converts units to meters and axes to Unity's Y-up, left-handed convention;
- welds duplicated vertices and rebuilds smooth normals with configurable hard edges;
- generates a level-of-detail (LOD) chain per part using quadric-error decimation;
- generates simplified physics collision meshes, separate from the render meshes;
- preserves the part hierarchy and part names of assemblies;
- attaches a `CADModelInfo` component with source metadata for twin tooling.

The package is pure C# — no native plugins — so editor imports work on any platform Unity's editor runs on, and the runtime loader ships in your builds unchanged.

2. Supported Formats

Format	Editor import	Runtime import	Notes
STL — binary & ASCII	Yes (drag & drop)	Yes	Multi-solid ASCII files import as separate, named parts. Facet normals are recomputed for smooth shading.
PLY — ascii, binary little/big endian	Yes (drag & drop)	Yes	Vertex colors and UVs are preserved; polygon faces are triangulated.
OBJ	Unity's native importer	Yes	The runtime loader supports groups, MTL diffuse colors, and negative indices.
STEP (.step / .stp)	Yes (via FreeCAD)	—	Assembly structure and part labels preserved. See section 7.
IGES (.iges / .igs)	Yes (via FreeCAD)	—	Same pipeline as STEP.

3. Installation

CAD Importer is distributed as a Unity package on GitHub, installed through the Unity Package Manager.

1. Open **Window** → **Package Manager**, press **+** → *Add package from git URL...* and paste:

```
https://github.com/Motawe3/CADImporter.git?path=Packages/com.motawea.cad-importer
```

To pin a specific release, append a tag, e.g. `#v1.0.0`.

2. Alternatively, add it to your project's `Packages/manifest.json`:

```
"com.motawea.cad-importer": "https://github.com/Motawe3/CADImporter.git?path=Packages/com.motawea.cad-importer#v1.0.0"
```

3. The package ships two assemblies — `CADImporter.Runtime` and `CADImporter.Editor`. No project settings are modified.
4. *Optional:* install [FreeCAD](#) (free, open source) to enable STEP/IGES import — see section 7.

Git requirement. Installing from a git URL requires a git client on your machine (git-scm.com); Unity's Package Manager invokes it during installation.

Render pipelines. Generated materials use *Universal Render Pipeline/Lit* and automatically fall back to *Standard* in Built-in render pipeline projects.

4. Importing in the Editor

4.1 Drag & drop

Drop an `.stl`, `.ply`, `.step` / `.stp` or `.iges` / `.igs` file anywhere under `Assets/`. The file imports like any model: the asset becomes a prefab containing the part hierarchy, meshes, materials, LOD groups and colliders as sub-assets. Drag it into a scene to use it.

4.2 Per-asset settings

Select the imported asset to edit its import settings in the Inspector (section 5), then press **Apply** to reimport. Settings are stored in the asset's `.meta` file, so they survive reimports and version control.

4.3 What gets generated

- **Root GameObject** named after the file, carrying `CADModelInfo`.
- **One child per part** (STL solid / PLY mesh / STEP solid), named from the source data.
- **LODGroup per part** (when enabled) with renderer children `Part_LOD0...N`.
- **Collider per part** according to the collider mode.
- **Mesh, material and collision-mesh sub-assets** inside the imported asset.

5. Import Settings Reference

Setting	Default	Description
Source Unit	Millimeters	Unit the file was authored in (mm, cm, m, inch, foot). Converted to meters — Unity physics and lighting assume meters.
Source Orientation	Z-up right-handed	Z-up RH matches SolidWorks, Inventor, CATIA, FreeCAD. Y-up RH matches most OBJ exports. Y-up LH means "already Unity", no conversion. Triangle winding is corrected automatically so surfaces stay outward-facing.
Additional Scale	1	Extra uniform scale on top of unit conversion.
Weld Vertices	On	Merges coincident vertices. STL stores three unique vertices per triangle; welding typically reduces vertex count ~6× and is required for LOD generation and smooth shading.
Weld Tolerance	1e-5 m	Merge distance, in meters, after scaling.
Force Recalculate Normals	Off	Recompute normals even when the file provides them. STL normals are always recomputed (per-facet normals shade poorly).
Smoothing Angle	30°	Edges sharper than this stay hard (crisp machined edges); flatter edges are smoothed with angle-weighted normals.
Generate LODs	On	Builds a LODGroup per part with decimated meshes.
LOD Qualities	0.5, 0.15	Triangle fraction kept at each extra LOD level. Two entries produce a three-level chain (100% / 50% / 15%).
LOD Min Triangles	512	Parts below this count skip LOD generation.
LOD First Transition	0.35	Screen height at which LOD0 hands over to LOD1. Remaining transitions are spaced geometrically down to the cull height.
LOD Cull Height	0.015	Screen height below which the part is culled.
Collider Mode	Simplified Mesh	None — attach your own. Box — bounds box, cheapest. Convex Mesh — convex hull, required for parts on dynamic Rigidbodies (robot links). Simplified Mesh — decimated non-convex mesh, best accuracy/cost trade-off for static geometry. Full Mesh — exact render mesh, most expensive.
Collider Quality	0.25	Triangle fraction kept in generated collision meshes.

Material Mode	Lit	Lit: URP Lit (or Standard). Vertex Color Unlit: shows per-vertex colors from PLY scan data.
Default Color	Light gray	Base color for parts without material data.
Read/Write Meshes	Off	Keep mesh data CPU-accessible at runtime. Off halves mesh memory.
Mark Static	Off	Flags objects for static batching/occlusion. Leave off for articulated robots; enable for factory shells and fixtures.
Generate Lightmap UVs	Off	Unwraps LOD0 meshes for baked lighting (slow on large models).
STEP Linear Deflection	0.1	Tessellation chord tolerance, relative to shape size. Lower = finer mesh, more triangles.
STEP Angular Deflection	20°	Tessellation angular tolerance. Lower = finer curvature.

6. The CAD Importer Window (Batch Import)

Open **Tools** → **CAD Importer**. The window lets you:

- queue external CAD files by drag & drop or the **Add Files...** button;
- edit one set of import settings applied to every queued file (persisted between sessions);
- configure the FreeCAD converter path for STEP/IGES;
- choose the target folder inside `Assets/`;
- optionally place an instance of each imported model in the open scene.

Press **Import**: each file is copied into the project, imported with your settings, and reported in the status box. From then on the copies behave like any other asset — adjust their settings individually in the Inspector at any time.

7. STEP / IGES Setup (FreeCAD)

STEP and IGES are B-rep (boundary representation) formats: they describe exact analytic surfaces, not triangles, and tessellating them requires a full CAD kernel. CAD Importer uses the Open CASCADE kernel bundled with **FreeCAD**, driven headlessly — FreeCAD never opens a window and is only executed during import.

1. Install FreeCAD 0.20 or newer from **freecad.org** (free, open source).
2. The importer auto-detects `FreeCADCmd.exe` in PATH and standard install folders. If detection fails, set the path in **Tools** → **CAD Importer** → **STEP / IGES Converter** → **Browse...**

3. Import `.step` / `.iges` files like any other asset.

Every solid in the file becomes a child `GameObject` named after its CAD label, so assemblies keep their structure — essential for mapping robot links or machine parts to simulation logic. Tessellation density is controlled by the two deflection settings (section 5).

Note. Without FreeCAD installed, STEP/IGES files import as an empty placeholder with a console warning explaining the requirement. All other formats work with no external dependencies.

8. Runtime Import for Digital Twins

Digital twins often need to load geometry that did not exist at build time. The `CADRuntimeImporter` API loads STL, PLY and OBJ files in the player, applying the same processing pipeline as the editor importers.

Demo sample. A ready-made scene ships with the package: open **Window** → **Package Manager** → **CAD Importer** → **Samples** and import *CAD Importer Demo* (requires URP). Its on-screen panel — the `DemoCadRuntimeImporter` component — imports files at runtime, or a generated sample part when the path field is left empty.

```
using CADImporter;
using UnityEngine;

public class TwinLoader : MonoBehaviour
{
    async void Start()
    {
        var settings = new CADRuntimeImportSettings
        {
            sourceUnit          = SourceUnit.Millimeters,
            sourceOrientation    = SourceOrientation.ZUpRightHanded,
            generateColliders    = true,
            colliderQuality      = 0.25f,    // decimated collision mesh
            convexColliders      = true      // required on dynamic Rigidbodies
        };

        // Parsing and geometry processing run on a worker thread;
        // only GameObject/Mesh creation touches the main thread.
        GameObject model = await CADRuntimeImporter.ImportAsync(
            @"C:\twins\gripper.stl", settings, transform);
    }
}
```

- `ImportAsync` keeps the simulation responsive; a synchronous `Import` overload exists for tooling.
- `CADRuntimeImportSettings.material` overrides the generated material.

- The returned root carries `CADModelInfo` (source file, format, unit, scale, triangle/vertex/part counts, import timestamp).

Shader inclusion in builds. Runtime-created materials use `Shader.Find("Universal Render Pipeline/Lit")`. Ensure that shader is included in builds — reference it in any scene material or add it to *Project Settings* → *Graphics* → *Always Included Shaders* — or supply your own material via the settings.

9. The Processing Pipeline & Performance

Each mesh passes through the following stages, in order:

1. **Unit scaling** — CAD millimeters become Unity meters. Correct scale is not cosmetic: PhysX solver behavior, gravity, and lighting all assume meters.
2. **Axis conversion** — Z-up right-handed data is mapped to Y-up left-handed with the triangle winding flipped, so normals keep pointing outward.
3. **Smooth normal generation** — angle-weighted normals across coincident positions; vertices split only where faces meet above the smoothing angle.
4. **Vertex welding** — duplicate vertices merged, degenerate triangles dropped, orphaned vertices pruned.
5. **LOD decimation** — a quadric-error-metric simplifier (Garland–Heckbert) builds each LOD level; borders are preserved and normals recomputed per level.
6. **Collision generation** — a separately decimated mesh feeds the collider, so physics never runs against render-resolution geometry.
7. **Unity mesh construction** — 16-bit index buffers whenever a part stays under 65k vertices; mesh data uploaded and released from CPU memory unless Read/Write is enabled.

Practical performance guidance

- Dozens of dense CAD assemblies in one scene are viable because distant parts render at 15% triangle density and cull entirely below 1.5% screen height.
- For robot links under dynamic Rigidbodies use **Convex Mesh** colliders — PhysX requires convex shapes on moving bodies.
- Keep **Read/Write Meshes** off unless a script needs vertex access; it halves runtime mesh memory.
- Static factory geometry: enable **Mark Static** and consider **Generate Lightmap UVs** for baked lighting.
- Extremely dense scan/STL data (millions of triangles): lower LOD0 cost by importing with a coarser STEP deflection or pre-decimating the source, then let the LOD chain handle the rest.

10. Working with Imported Models in Robotics Scenes

- **Articulation:** imported parts are plain GameObjects — add `ArticulationBody` or `Rigidbody` components to the parts that move, and parent them according to your kinematic chain. Part names from the CAD assembly make this mapping straightforward.
- **Metadata:** `CADModelInfo` on every imported root records the source file, format, unit and counts — useful to keep twins traceable to their CAD source.
- **Re-import on change:** replacing the CAD file on disk reimports with the same settings; prefab instances in scenes update automatically.
- **Collision layers:** generated colliders live on the part GameObjects, so standard layer/matrix configuration applies unchanged.

11. Troubleshooting & FAQ

The model imports tiny / huge

Set **Source Unit** to match the exporting CAD package (usually millimeters), Apply.

The model lies on its side

Set **Source Orientation** to *Z-up right-handed* (typical CAD) or *Y-up right-handed* (typical OBJ/DCC exports).

Surfaces look faceted or too smooth

Adjust **Smoothing Angle**: raise it to smooth more, lower it to keep edges hard. For files with authored normals, try **Force Recalculate Normals**.

STEP file imports as an empty object

FreeCAD is missing or not detected — see section 7. The console warning contains the same instructions.

Import is slow on a huge STL

Most time goes to welding and LOD decimation. Reduce LOD levels, raise **LOD Min Triangles**, or disable **Generate Lightmap UVs**.

A part flickers between LOD levels

Raise **LOD First Transition** or reduce the number of LOD qualities for that asset.

PLY vertex colors don't show

Set **Material Mode** to *Vertex Color Unlit*, or use your own vertex-color shader; the colors are stored in the mesh either way.

12. Known Limitations

- STEP/IGES import requires FreeCAD and is editor-only (no runtime B-rep tessellation).
- Decimated LOD meshes drop vertex colors and UVs (rare in CAD data); normals are recomputed.
- Nonstandard STL per-facet color extensions (SolidWorks/Magics attribute bytes) are ignored.
- OBJ assets in the editor use Unity's native importer; CAD Importer settings apply to OBJ only through the runtime API.
- STEP assembly transforms are baked into part geometry (parts arrive positioned, not as reusable instances).

13. Support

Questions, bug reports and feature requests: **github.com/Motawe3/CADImporter/issues** (preferred) or **mohammed.motawea90@gmail.com** — please include the Unity version, the importer settings used, and (when possible) a sample file that reproduces the issue.

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